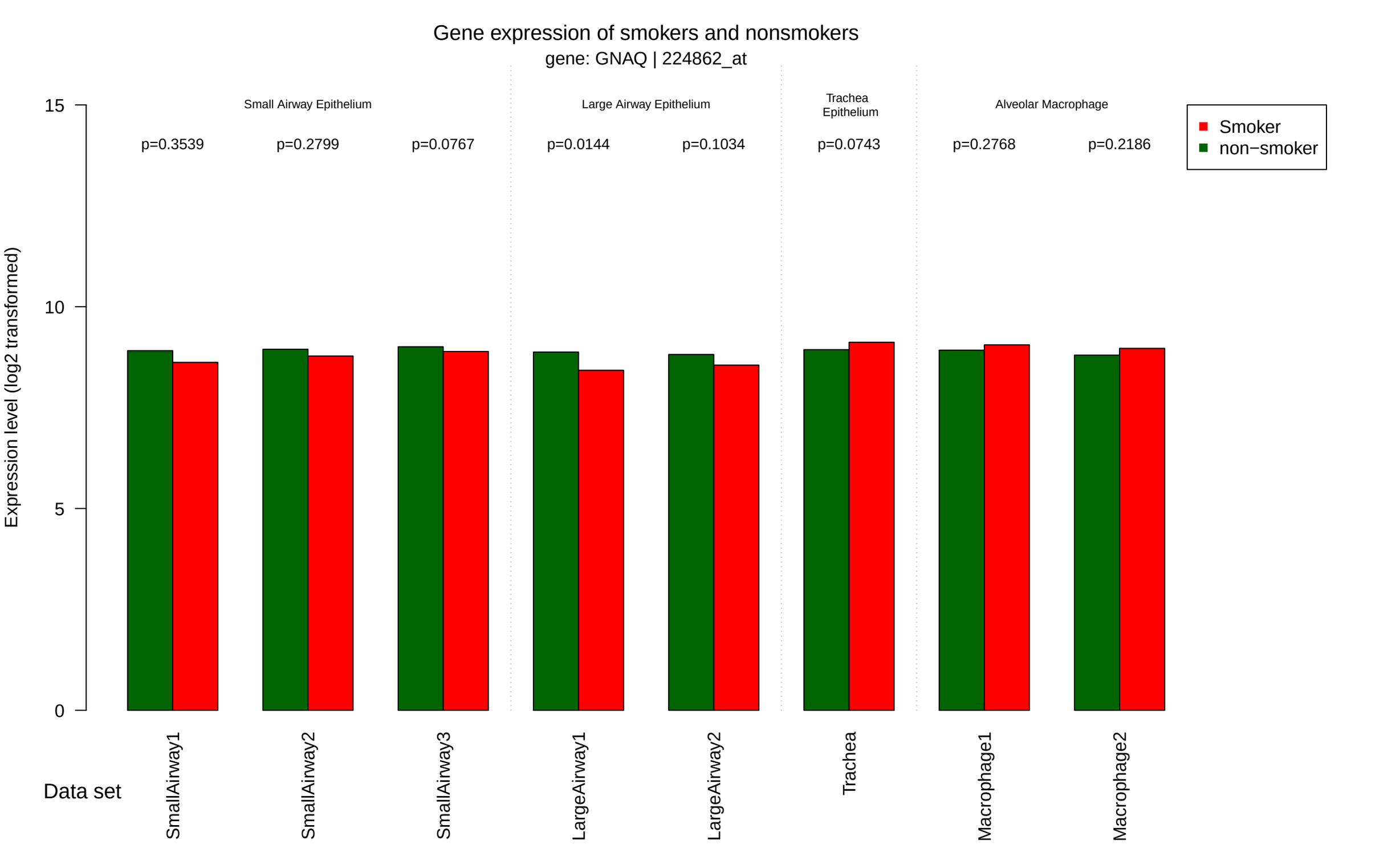
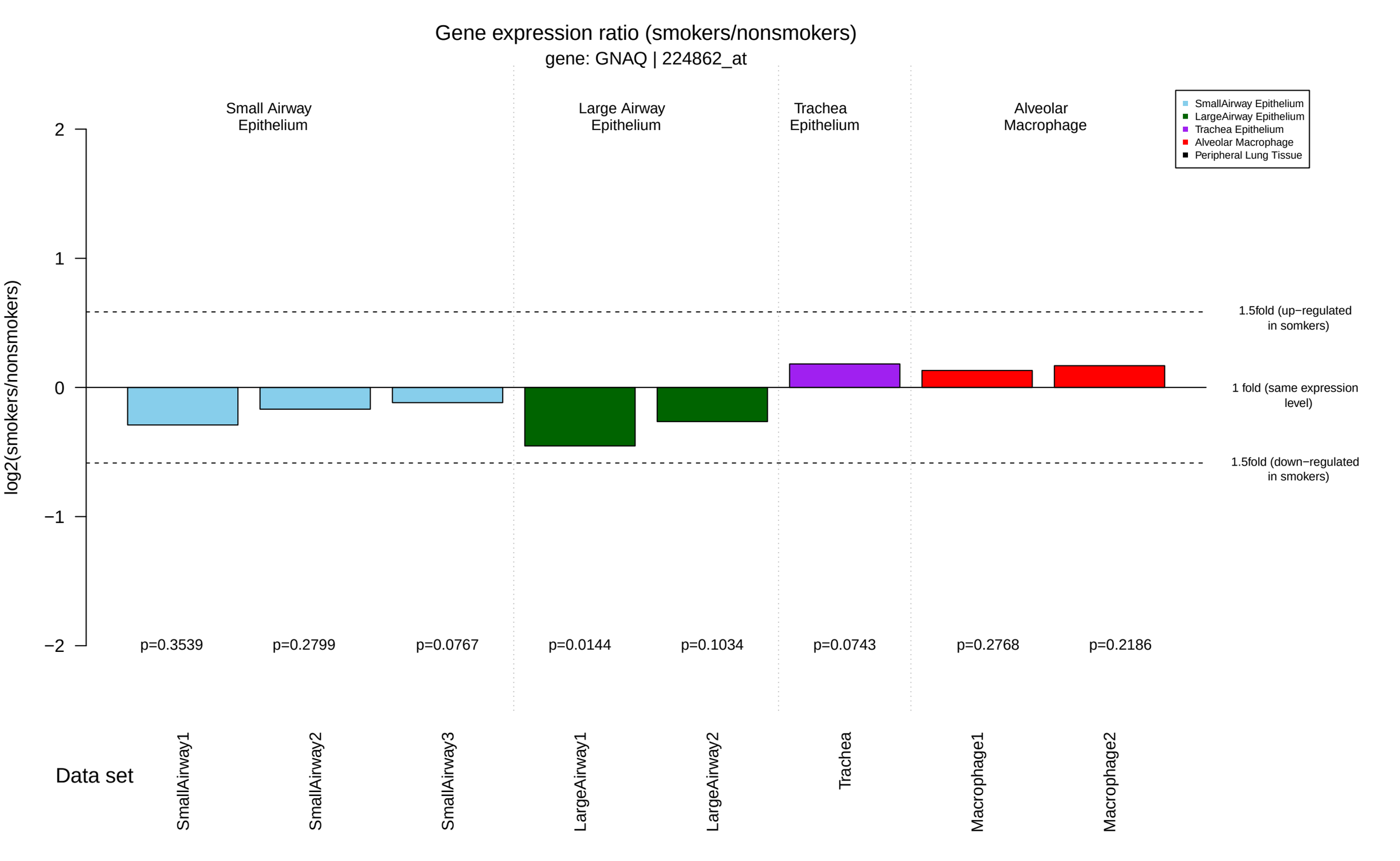
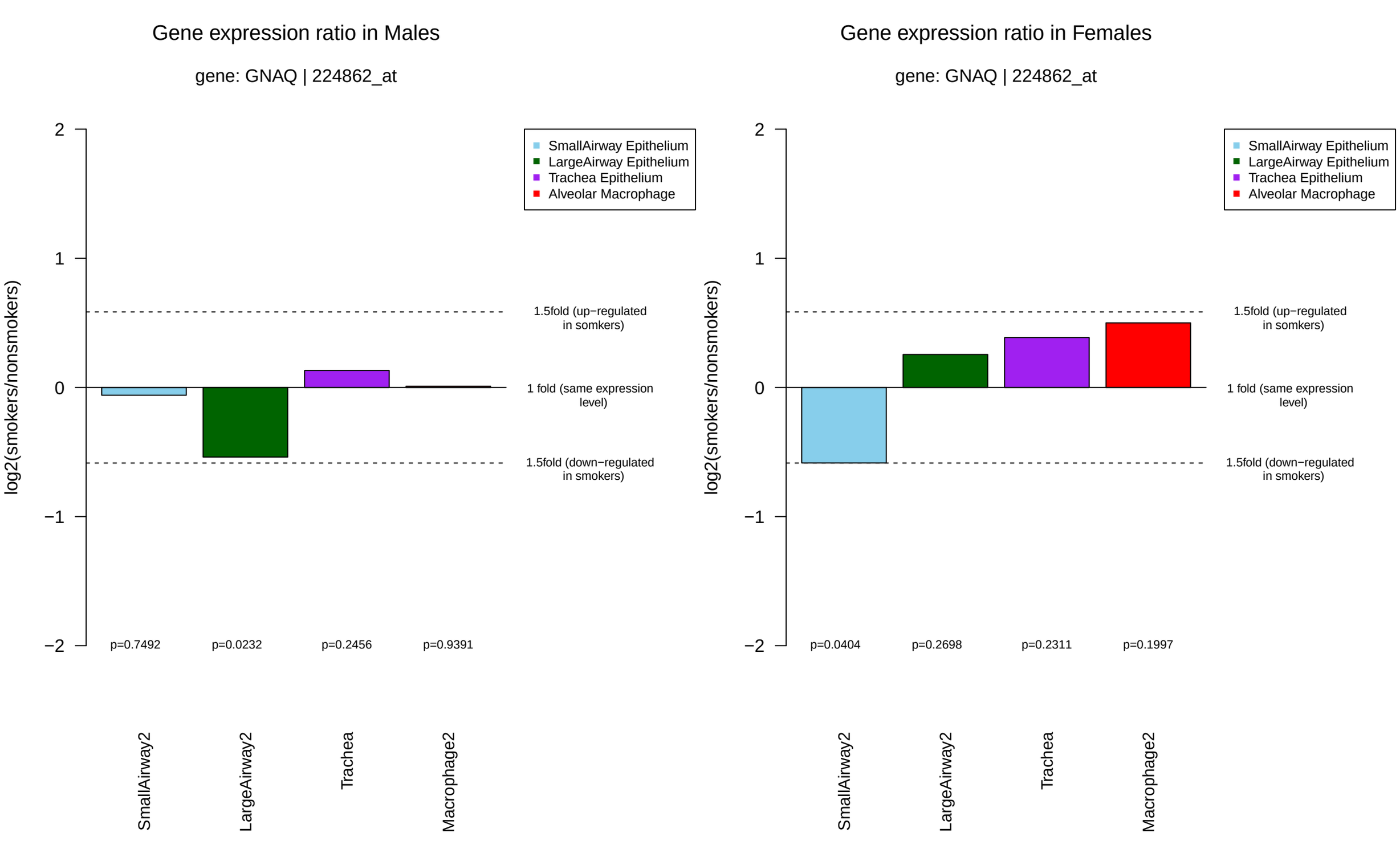
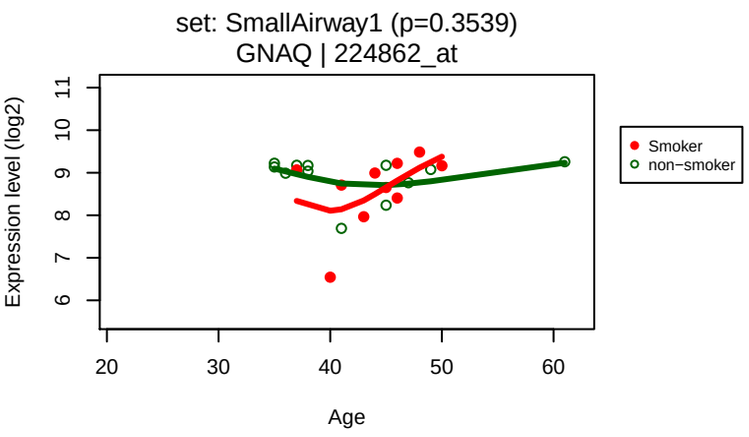
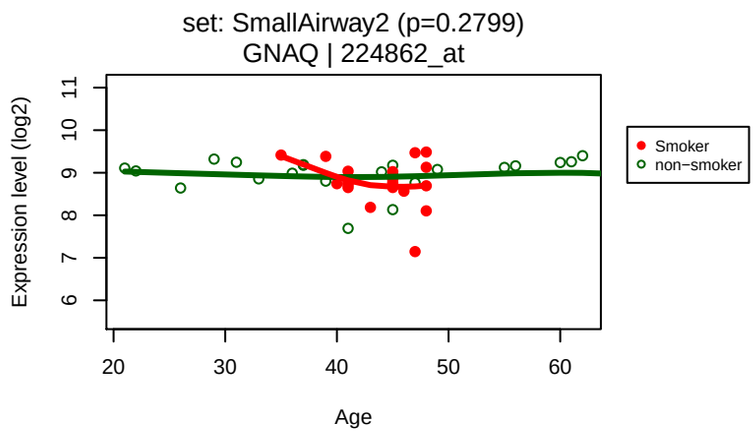
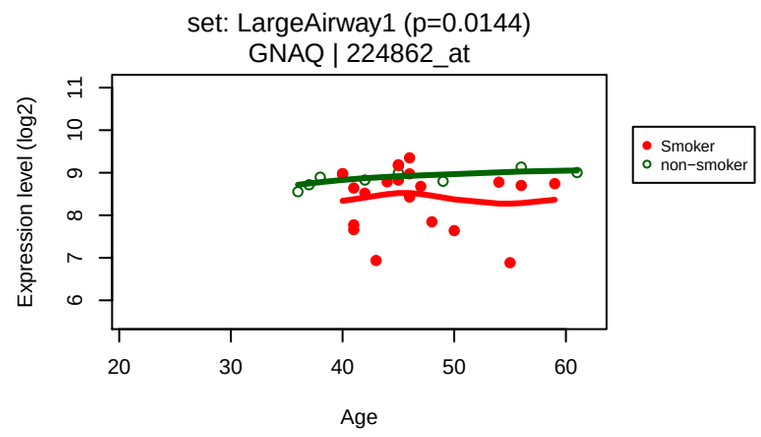

Gene expression of smokers and nonsmokers
gene: GNAQ | 224862_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3539
p=0.2799
p=0.0767
p=0.0144
p=0.1034
p=0.0743
p=0.2768
p=0.2186
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

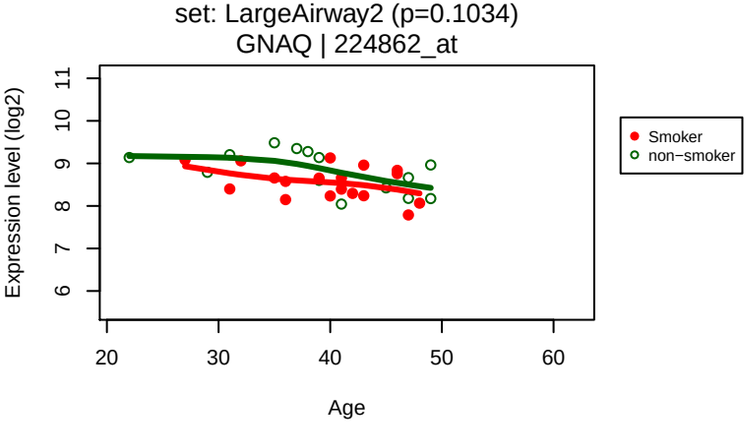

Gene expression ratio (smokers/nonsmokers)
gene: GNAQ | 224862_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3539
p=0.2799
p=0.0767
p=0.0144
p=0.1034
p=0.0743
p=0.2768
p=0.2186
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

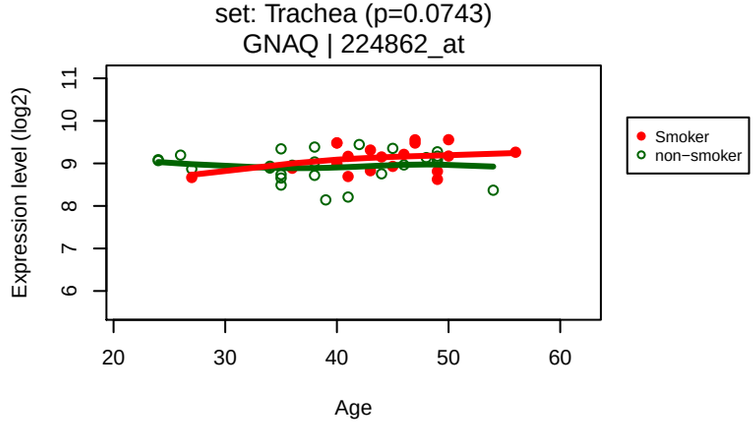

Gene expression ratio in Males
gene: GNAQ | 224862_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7492
p=0.0232
p=0.2456
p=0.9391
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAQ | 224862_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0404
p=0.2698
p=0.2311
p=0.1997
SmallAirway2
LargeAirway2
Trachea
Macrophage2

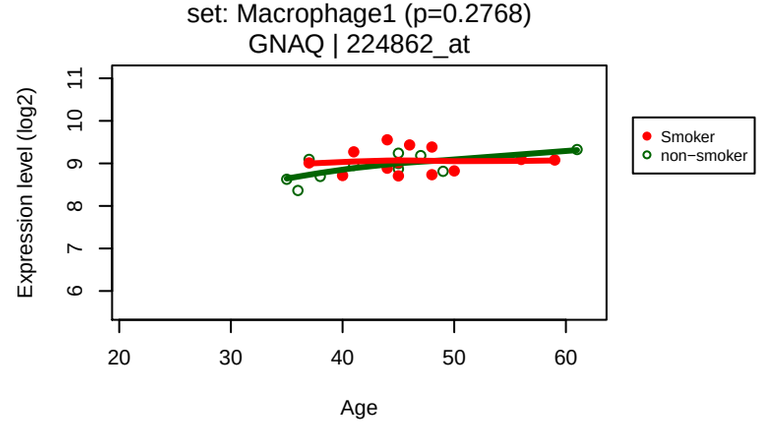

set: SmallAirway1 (p=0.3539)
GNAQ | 224862_at
Expression level (log2)
Age
Smoker
non−smoker

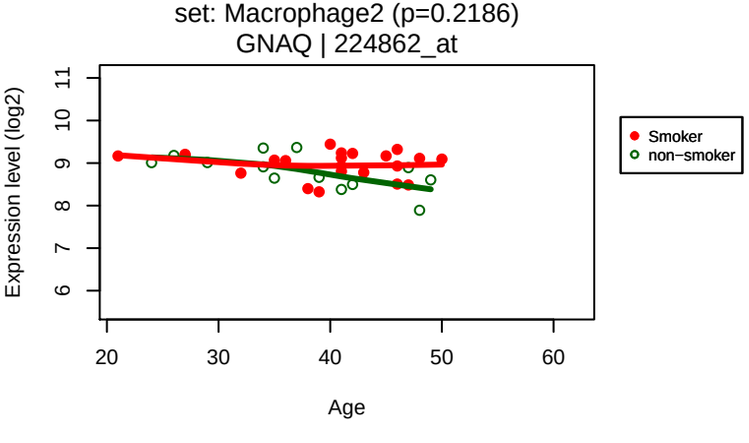

set: SmallAirway2 (p=0.2799)
GNAQ | 224862_at
Expression level (log2)
Age
Smoker
non−smoker

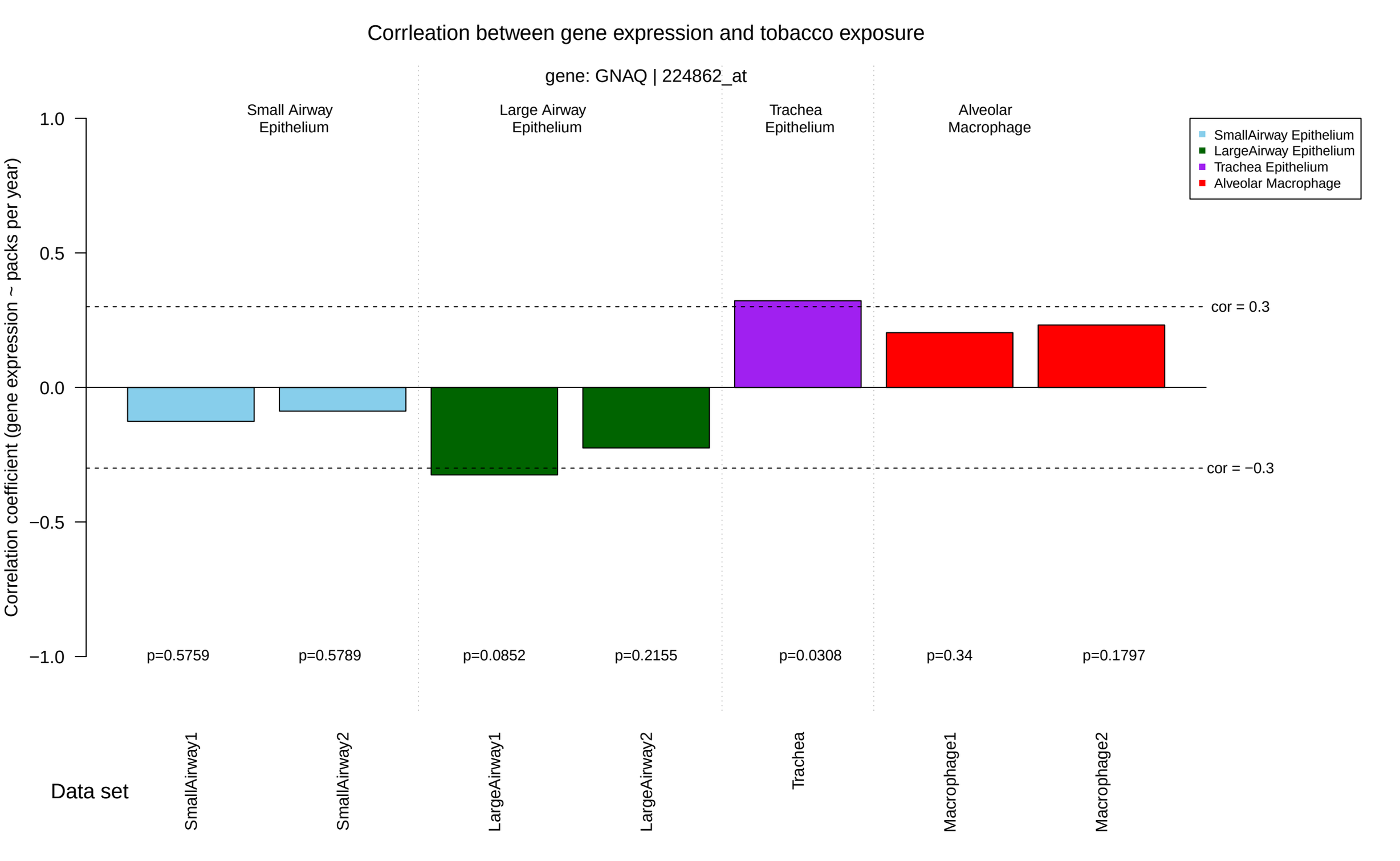

set: LargeAirway1 (p=0.0144)
GNAQ | 224862_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1034)
GNAQ | 224862_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0743)
GNAQ | 224862_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2768)
GNAQ | 224862_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2186)
GNAQ | 224862_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAQ | 224862_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5759
p=0.5789
p=0.0852
p=0.2155
p=0.0308
p=0.34
p=0.1797
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2